

ES647 Term Paper Part I

Part I. Composition and Characterisation of MSW

For the town of Aligarh, the composition of Municipal Solid Waste is found as follows - (Take 100 kgs of MSW)

S.No.	Components	Wet Mass (kg)	Moisture Content (%)	Dry Mass (kg)
1.	Organic Waste	55.5	50	27.75
2.	Paper	12	6	11.28
3.	Plastic	6.2	2	6.076
4.	Metals	1.1	3	1.067
5.	Glass	0.7	2	0.686
6.	Wood	0.6	20	0.48
7.	Textiles	2.7	10	2.43
8.	Inert	21.2	8	19.504
		100		69.273

$$a) \text{ Moisture Content (Wet basis)} = \frac{100 - 69.273}{100} \times 100 = 30.727\%$$

$$\text{Moisture Content (Dry mass basis)} = \frac{100 - 69.273}{69.273} \times 100 = 44.356\%$$

B) For Density Calculations, we will calculate Volume equivalent to the solid waste.

Component	Wet Mass (kg)	Typical Density (kg/m ³)	Volume (m ³)
Organic Waste	55.5	240	0.231
Paper	12	85	0.141
Plastic	6.2	65	0.095
Metals	1.1	200	0.005
Glass	0.7	195	0.004
Wood	0.6	240	0.002
Textiles	2.7	65	0.042
Inert	21.2	480	0.044
			0.564

$$\text{Density of solid waste} = \frac{100}{0.564} = 177.3 \text{ kg/m}^3$$

c)

Components	Wet Mass (kg)	Dry Mass (kg)	Composition					
			C	H	O	N	S	Ash
Organic Waste	55.5	27.75	13.32	1.78	10.43	0.72	0.11	1.39
Paper	12	11.28	4.91	0.68	4.96	0.03	0.02	0.68
Plastics	6.2	6.08	3.65	0.44	1.39	---	---	0.60
Wood	0.6	0.48	0.24	0.03	0.2	---	---	0.01
Textiles	2.7	2.43	1.34	0.16	0.76	0.11	---	0.06
Inert	21.2	19.5	5.13	0.58	0.39	0.10	0.04	13.26
Total	98.2	67.52	28.59	3.67	18.13	0.96	0.17	16.0

For chemical Formula without water,

Component	Mass (kg)	Rg/mole	Moles	Ratio with S	Ratio withouts
C	28.59	12	2.382	476.4	34.5
H	3.67	1	3.67	734.0	53.2
O	18.13	16	1.133	226.6	16.4
N	0.96	14	0.069	13.8	1.0
S	0.17	32	0.005	1.0	—

Chemical formula without water

With S $\rightarrow C_{476.4} H_{734} O_{226.6} N_{13.8} S$

Without S $\rightarrow C_{34.5} H_{53.2} O_{16.4} N$

For chemical formula with water, add contribution of water from moisture content.

$$\text{Moisture} = 98.2 - 67.52 = 30.68 \text{ kg}$$

Components	Mass (kg)	kg/mole	Moles	Ratio with S	Ratio without S
C	28.59	12	2.382	476.4	34.5
H	$3.67 + \frac{30.68 \times 2}{18} = 7.08$	1	7.08	1416	102.6
O	$18.13 + \frac{30.68 \times 16}{18} = 45.4$	16	2.837	567.4	41.1
N	0.96	14	0.069	13.8	1
S	0.17	32	0.005	1	—

Chemical formula with water

With S $\rightarrow C_{476.4} H_{1416} O_{567.4} N_{13.8} S$

Without S $\rightarrow C_{34.5} H_{102.6} O_{41.1} N$

d)

Components	Wet Mass (kg)	Typical Energy Content (MJ/kg)	Total Energy as discarded (MJ/kg)
Organic Waste	55.5	18.00	999
Paper	12	16.75	201
Plastics	6.2	32.6	202.12
Glass	0.7	0.15	0.105
Wood	0.6	18.6	11.16
Textile	2.7	17.45	47.115
Inert	21.2	7.00	148.4
			1608.9

$$\begin{aligned} \text{Energy Content (as discarded)} &= \frac{1608.9}{100} \\ &= 16.089 \text{ MJ/kg} \end{aligned}$$

$$\text{Moisture Content} = 30.727\%$$

$$\begin{aligned}\text{Energy Content (Dry Basis)} &= 16.089 \times \frac{100}{100 - 30.727} \\ &= 23.225 \text{ MJ/kg}\end{aligned}$$

Assume Ash Content = 20%

$$\begin{aligned}\text{Energy Content on ash free dry basis} &= 16.089 \times \frac{100}{100 - 20 - 30.727} \\ &= 32.65 \text{ MJ/kg}\end{aligned}$$

Based on Du long Formula,

$$\text{Energy Content}_{HV} (\text{BTU/lb}) = 145C + 620\left(H - \frac{O}{8}\right) + 40S$$

→ C, S, H, O are equivalent weight of SW

$$= 145 \times 28.59 + 620\left(3.67 - \frac{18.13}{8}\right) + 40 \times 0.17$$

$$= 4145.55 + 870.325 + 6.8$$

$$\begin{aligned}&= 5022.7 \text{ BTU/lb} \quad \left(1 \text{ BTU/lb} = 2.326 \text{ kJ/kg}\right) \\ &= 11,682.8 \text{ kJ/kg}\end{aligned}$$

$$\text{HV (MJ/kg)} = 11.68 \text{ MJ/kg}$$